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9th International Mathematics Assessments for Schools (2019-2020)

Middle Primary Division Round 2

Time: 120 minutes

Printed Name _____ *Code* _____ *Score* _____

Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the space provided. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the space provided to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

The following area is to be filled in by the judges;
the contestants are not supposed to mark anything here.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total Score	Signature
Score																	
Score																	

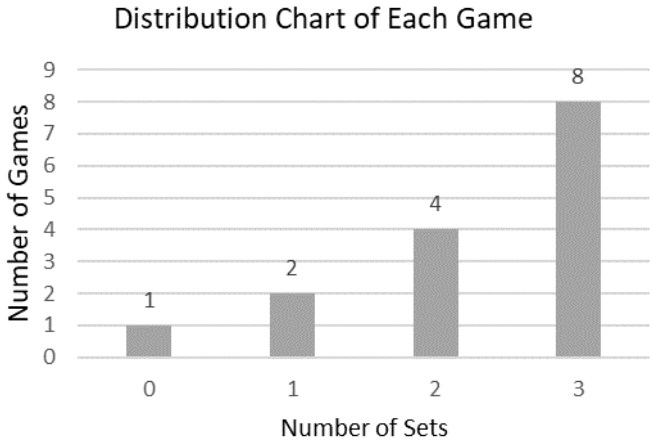
Middle Primary Division Round 2

Questions 1 to 5, 4 marks each

1. Divide the number 10 into the sum of three distinct positive integers. How many different ways are there altogether? (Note: if the arrangement of same three positive integers are in different order, they are regarded as the same).
（A）1　　（B）2　　（C）3　　（D）4　　（E）5

Answer：_____

2. In a volleyball tournament, each game adopts the best three sets out of five sets game, in which each team has 0, 1, 2, or 3 possible sets of winning. The distribution chart shows a number of sets that a team can win as follows.



What is the median of the number of sets won by the team in the tournament? (Note: The median of a group of data is the middle datum when the data are arranged in ascending order of magnitude.)
（A）1　　（B）1.5　　（C）2　　（D）2.5　　（E）3

Answer：_____

3. In the following addition expression, the same symbol represents the same number, and different symbols represent different numbers. It is known the leftmost digit in both the addend and the sum is nonzero. What is the value of △+□+○?

$$\begin{array}{cccc} & ○ & 8 & ○ \\ + & △ & □ & ○ \\ \hline △ & □ & ○ & 8 \end{array}$$

（A）10　　（B）11　　（C）12　　（D）13　　（E）14

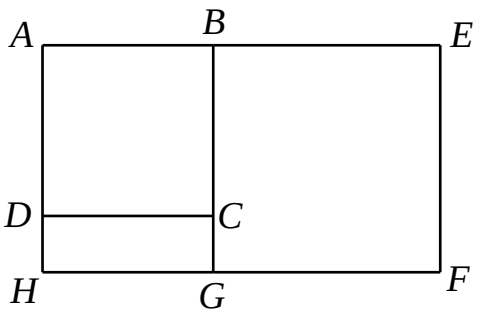
Answer：_____

4. Allan and Benny are computing the sum of a three-digit number and a two-digit number. The computation of Allan is correct with an answer of 136, while Benny is not doing it correctly because of a misalignment of digits in the vertical format mistakenly adding the tens digit of three-digit number with the units digit of the two-digit number and obtaining the tens digit in the sum. This is followed by adding the hundreds digit of the three-digit number and the tens digit of the two-digit number to get the hundreds digit in the sum with the final answer becoming 244. What is the difference between the three-digit number and the two-digit number?

（A）106　（B）108　（C）110　（D）112　（E）114

Answer：_____

5. As shown in the figure, both quadrilateral *ABCD* and quadrilateral *BEFG* are squares, and $AE = 28$ cm, $AH = 16$ cm. What is the perimeter, in cm, of the rectangle *CDHG*?



（A）20　（B）24　（C）32　（D）48　（E）64

Answer：_____

Questions 6 to 13, 5 marks each

6. There are 45 students in the class sitting in a row. The position of the student A is 28th from left, and the position of student B is 35th from right. How many students are there between student A and student B?

Answer：_____ students

7. There are 14 rows of seats in a semi-circular auditorium. In a certain performance, 18 spectators are seated at the first row, and, starting from the second row, each row has three more spectators than the previous one. How many spectators are there in the auditorium during the performance?



Answer： _____ spectators

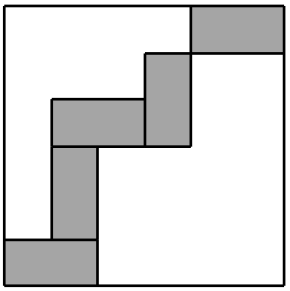
8. What is the value of $(1\times2\times3\times\cdots\times11)\div(22\times24\times25\times27)$? (if the answer is a fraction, express the answer with simplest form)

Answer： _____

9. In a certain examination, Lily's average score in the three subjects - Foreign Language, Mathematics and Nature Science - was 90 points while the English examination still needs to be scheduled to a further date. How many points must Lily need to have in English to make the average score of the four subjects to be 92 points?

Answer： _____ points

10. As shown in the figure, place five congruent small rectangles inside a large square such that each side of all the rectangles is parallel to one of the sides of the square. If the side length of the given large square is 48 cm, what is the area, in cm^2, of a small rectangle?

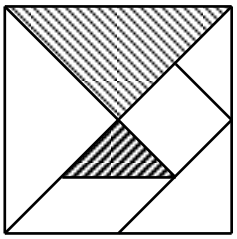


Answer： _____ cm^2

11. An electronic watch shows 7 o'clock thirty-one minutes on the surface of the watch as 7:31. From 7:00 to 8:00 in the morning, how many times will three distinct digits appear as a point of time on the surface of watch?

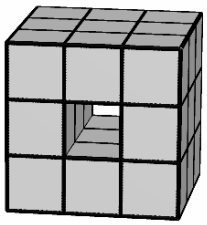
Answer： _____ times

12. Below is a tangram which was invented in China. It is a famous puzzle with seven ingenious pieces: five isosceles right triangles of three sizes, one square and one parallelogram. If the area of one of the smallest isosceles right triangle in the 7-piece puzzle is 6 cm^2, what is the area, in cm^2, of the largest isosceles right triangle in this puzzle?



Answer : _____ cm^2

13. There are 24 regular unit cube blocks glued together as shown in the figure below. How many faces of regular unit cubes blocks are glued together?



Answer : _____ faces

Questions 14 to 15, 20 marks each
(Detailed solutions are needed for these two problems)

14. After arranging 1000 consecutive positive integers in a certain order, followed by computing the sum of each successive three numbers from the beginning, a total of 998 sums can be obtained. What is the largest possible number of odd integers among these 998 sums?

Answer : _____ odd integers

15. Fill in the grid with the positive integers starting from 1 as shown in the figure, and then use an "L-shaped block" (that is composed of three unit squares which can be rotated or flipped) to frame three numbers and compute their sum.
For example, in the figure below, the sum of the three numbers framed by the L-shaped block is $10+11+18=39$. It is known that the sum of the three numbers framed by an L-shaped block is 2019. What is the largest possible value of the largest number among the three numbers framed by an L-shaped block?

1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
…	…	…	…	…	…	…

Answer：_____